

# INTERNATIONAL CYBERSECURITY AND DIGITAL FORENSICS ACADEMY

Nasarawa State, Nigeria

## ACICP 101 — Physics & Electrical Circuit Fundamentals for ICS/SCADA

### Week 1 Lab Safety Readiness Pack

*Low-voltage electrical safety, lab discipline, risk assessment and readiness declaration*

|                             |                                                           |
|-----------------------------|-----------------------------------------------------------|
| <b>Course Code:</b>         | ACICP 101                                                 |
| <b>Lab Title:</b>           | Week 1 Lab Safety Readiness Pack                          |
| <b>Student Name:</b>        | SUNUSI ABDULLAHI                                          |
| <b>Registration Number:</b> | C11/26/ACIS/17362                                         |
| <b>Department:</b>          | Cybersecurity (ACIS Programme)                            |
| <b>Assessment Type:</b>     | Lab safety readiness, hazard identification, pre-lab quiz |
| <b>Due Date:</b>            | Friday, 03 Jul 2026, 11:59 PM WAT                         |
| <b>Date of Submission:</b>  | 02/JULY/2026                                              |
| <b>Version:</b>             | 1.0                                                       |

## Executive Summary

---

This document is the Week 1 Lab Safety Readiness Pack submission for ACICP 101. It covers three sections: Part A is a personal readiness checklist confirming that ten basic lab safety competencies are understood and can be applied before any circuit work begins. Part B is a hazard identification table that identifies the hazard, possible consequence and one practical control measure for five realistic low-voltage scenarios. Part C is a pre-lab knowledge quiz of ten questions, including one Ohm's Law calculation shown in full with formula, substitution and answer. All answers are written plainly and honestly as required by the submission guidelines. This pack demonstrates that the student is ready to begin safe electrical circuit work in the ACICP 101 virtual and supervised training environment.

### Part A: Lab Safety Readiness Checklist

---

*Instructions: For each item below, the student provides a short written response or evidence statement confirming the readiness requirement is understood. Each item is worth 1 mark.*

| No. | Readiness Requirement                                                  | Student Response / Evidence                                                                                                                                                                                                                                                                                                                      | Marks |
|-----|------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 1   | I can identify the power source rating before connecting any circuit.  | I check the label on the supply unit or the circuit diagram for the rated voltage and current output before connecting any load. For example, a bench supply labelled 24 V DC / 2 A means it can deliver a maximum of 2 A. I calculate the expected load current using $I = V / R$ and confirm it is within the supply rating before connecting. | 1     |
| 2   | I can explain why circuits must be switched off before wiring changes. | If the circuit is live when a wire is moved or added, the exposed conductor can touch another terminal or wire and create a short circuit, a spark, or an electric shock. Switching the supply off before any wiring change removes all voltage from the circuit and makes it safe to handle the components and conductors.                      | 1     |
| 3   | I can identify polarity on DC power supply terminals.                  | On a DC supply, the positive terminal is marked with the + symbol (and is often coloured red), and the negative terminal is marked with the - symbol (and is often coloured black or has a COM label). On a circuit diagram the longer line of the battery symbol is positive and the shorter line is negative. I                                | 1     |

|   |                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
|---|-----------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
|   |                                                                       | always connect positive to positive and negative to negative before energising.                                                                                                                                                                                                                                                                                                                                                              |   |
| 4 | I can calculate expected current before connecting a load.            | I use Ohm's Law: $I = V / R$ . For example, a 160 ohm load on a 24 V supply draws $I = 24 / 160 = 0.15 \text{ A} = 150 \text{ mA}$ . I calculate this first and confirm that neither the supply nor the load rating is exceeded before connecting anything.                                                                                                                                                                                  | 1 |
| 5 | I know how to avoid short circuits across supply terminals.           | A short circuit happens when a low-resistance or zero-resistance path is connected directly across the supply. I avoid this by: (1) never placing an unprotected wire from the + terminal directly to the - terminal without a load in series, (2) always including a fuse or current-limiting resistor in the circuit, and (3) routing wires carefully so that no bare conductor touches the opposite terminal.                             | 1 |
| 6 | I can identify hot components caused by excess power dissipation.     | A component becomes hot when it is dissipating more power than it is rated for, according to $P = I^2 \times R$ . Signs of excess dissipation include a warm or burning smell, discolouration or brown marks on a resistor, or smoke. I detect this by observing the component carefully after energising. If any component is unusually warm or smells, I switch off immediately and re-check the circuit values against component ratings. | 1 |
| 7 | I understand why fuses and current limits protect wiring and devices. | A fuse is a deliberate weak link in the circuit. If current exceeds the fuse rating, the fuse wire melts and opens the circuit before the wiring or devices are damaged by heat. Without a fuse, excess current would overheat and melt the wire insulation, potentially causing a fire or destroying connected equipment. Current-limit settings on a bench supply serve the same protective role electronically.                           | 1 |
| 8 | I can keep liquid, metal tools and loose wires away from the circuit. | Liquids are electrically conductive and can create unexpected current paths across terminals, damaging components or causing a shock. Metal tools left across terminals cause short circuits. Loose wires can fall across the wrong terminals during vibration or movement. I ensure the work area is dry and clear before energising, tools are stored away from the circuit board, and all free wire ends are insulated or secured.        | 1 |

|    |                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
|----|-------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| 9  | I know how to stop work and report unsafe conditions.                   | If I observe smoke, unusual heat, burning smell, sparking, a tripped breaker, or any condition that feels unsafe, I immediately switch off the supply using the main power switch, do not touch the circuit, inform the instructor or lab supervisor, and wait for clearance before resuming. I do not attempt to investigate a live fault alone.                                                                                        | 1 |
| 10 | I understand that mains-voltage work is not permitted in this activity. | This course uses only low-voltage DC sources: battery packs, USB-powered trainers, or protected 5 V / 12 V / 24 V DC bench supplies under supervision. Connection to public mains electricity (230 V AC in Nigeria) is strictly prohibited in this activity. Any activity involving AC mains, motors, industrial panels or field equipment must be demonstrated by a qualified instructor only. I will not attempt any mains connection. | 1 |

## Part B: Hazard Identification Table

*Instructions: For each scenario, identify the possible hazard, describe the possible consequence, and state one practical control measure.*

| Scenario                                                               | Possible Hazard                                                                         | Possible Consequence                                                                                                                                                        | Control Measure                                                                                                                                                                                                                     | Marks |
|------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <b>Connecting a 24 V supply directly across a wire with no load</b>    | Short circuit — the wire has near-zero resistance, so an extremely large current flows. | The wire overheats and melts its insulation, the supply is overloaded and may be permanently damaged, and there is a risk of fire or burns to the person handling the wire. | Always connect a load (resistor, relay coil, sensor) in series before energising. Calculate the expected current first using $I = V / R$ to confirm it is within safe limits. Include a fuse rated just above the expected current. | 2     |
| <b>Using a resistor with too low wattage in a high-current circuit</b> | Thermal overload — the resistor dissipates more power than its rated                    | The resistor overheats, may crack, burn, catch fire, or produce toxic fumes. It may also fail open-circuit or short-circuit, disrupting the                                 | Calculate expected power before selecting the component: $P = I^2 \times R$ . Choose a resistor with a wattage rating at least double                                                                                               | 2     |

|                                                                                  |                                                                                                                                             |                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                       |   |
|----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
|                                                                                  | wattage, $P = I^2 \times R$ .                                                                                                               | rest of the circuit and potentially damaging other components.                                                                                                                                                                                                 | the calculated value to give a safety margin. For example, if calculated $P = 0.3 \text{ W}$ , use a 1 W rated resistor.                                                                                                                                                              |   |
| <b>Reversing polarity on an electrolytic capacitor or DC device</b>              | Reverse-polarity damage — electrolytic capacitors and many DC devices (diodes, transistors, PLC inputs) are designed for one polarity only. | An electrolytic capacitor connected with reversed polarity can vent, leak electrolyte, or rupture violently. DC devices such as a PLC analogue input card connected in reverse polarity may be permanently destroyed and the supply may be short-circuited.    | Before energising, verify polarity on every component using the supply markings (+/-), the component data sheet, or a multimeter set to DC voltage. Connect the positive supply rail to all marked + terminals and 0 V rail to all marked - terminals before switching on.            | 2 |
| <b>Changing circuit connections while power is still ON</b>                      | Live-circuit shock or short circuit — bare conductors are exposed during the connection change while voltage is present.                    | The person making the change can receive an electric shock from the live terminal. A bare wire touched to the wrong terminal creates a short circuit, which can blow a fuse, damage the supply, overheat wires, or destroy connected components in an instant. | Always switch the supply OFF before making any wiring change. After the change is complete and all connections are confirmed correct, switch the supply back ON. This is a fundamental electrical safety rule: de-energise before touching any conductor.                             | 2 |
| <b>Sharing one power source across several unknown loads without calculation</b> | Supply overload — the total current drawn by all unknown loads together may exceed the supply's rated output current.                       | The supply overheats and shuts down, its output voltage drops or becomes unstable, causing all connected devices to malfunction or receive incorrect voltage. In the worst case the supply is permanently damaged or, if                                       | Calculate the total expected current of all loads before connecting: $I(\text{total}) = I_1 + I_2 + I_3 + \dots$ . Confirm this total is below the supply's rated output current. If it exceeds the rating, use a larger supply or split the loads across two separate power sources. | 2 |

|  |  |                                                                     |  |  |
|--|--|---------------------------------------------------------------------|--|--|
|  |  | unprotected, the supply wiring overheats and creates a fire hazard. |  |  |
|--|--|---------------------------------------------------------------------|--|--|

## Part C: Pre-Lab Knowledge Quiz

---

*Instructions: Answer all ten questions. Question 1 requires a full calculation showing formula, substitution and answer with unit. All other questions require a clear written explanation.*

**Q1. A 24 V DC supply is connected to a 120 ohm relay coil. Calculate the expected current.**

Given:  $V = 24 \text{ V}$ ,  $R = 120 \text{ ohm}$

Formula:  $I = V / R$

Substitute:  $I = 24 / 120$

Answer:  $I = 0.2 \text{ A} = 200 \text{ mA}$

**Q2. A student measured 2.4 A in a small resistor circuit expected to draw 0.24 A. State two possible faults.**

Fault 1: The resistor value used is ten times smaller than the intended value (for example, a 100 ohm resistor was connected where a 1000 ohm resistor was required). By Ohm's Law, a ten-times smaller resistance draws ten-times more current.

Fault 2: There is a partial short circuit in the circuit — an unintended low-resistance connection is bypassing some or all of the resistor, reducing the total circuit resistance by a factor of ten and causing the current to rise to 2.4 A.

**Q3. Why is it unsafe to bypass a fuse during testing?**

A fuse is the only component that protects the wiring and devices against excessive current. If the fuse is bypassed (replaced with a wire or a piece of metal), there is nothing to interrupt the circuit if a fault causes an overcurrent. The wiring will overheat and melt its insulation, connected components will be destroyed, and there is a serious risk of fire. The fuse must always remain in place during testing. If a fuse keeps blowing, the underlying fault must be found and corrected, not bypassed.

**Q4. What is the difference between a short circuit and an open circuit?**

A short circuit is a zero-resistance or very low-resistance path that connects two points in the circuit that should not be directly connected (usually the + and - supply rails). It causes an extremely large current to flow, overheating wires and damaging the supply. A short circuit is a low-impedance fault.

An open circuit is the opposite — it is a break or disconnection in the circuit path where there should be a continuous conductor. No current can flow at all through an open circuit. In ICS/SCADA, a broken sensor wire or a blown fuse both create an open circuit, causing the signal to be lost or the device to stop working. An open circuit is a high-impedance fault.

**Q5. A 0.25 W resistor is dissipating 0.40 W. Explain what may happen and what should be changed.**

The resistor is operating at 0.40 W, which is 60% above its rated maximum of 0.25 W. This means the resistor is absorbing more heat energy than it can safely release. What may happen: the resistor body temperature rises above its safe limit, causing it to burn, discolour, crack, produce smoke, or fail permanently (either open-circuit or short-circuit). What should be changed: the resistor must be replaced with one rated for a higher wattage. The minimum safe replacement is at least double the dissipated power — so a 1 W rated resistor should be used in place of the 0.25 W part. Alternatively, if the current in the circuit is too high, the circuit design must be corrected to reduce power to within the component rating.

**Q6. State two reasons why a student must document circuit values before energising the circuit.**

Reason 1: Documentation of voltage, resistance and expected current allows the student to check in advance that no component rating will be exceeded. This prevents damage to components and avoids safety hazards such as overheating resistors or overloaded supplies.

Reason 2: If the circuit behaves unexpectedly after energising (wrong current, wrong voltage, component failure), the pre-recorded values give a baseline to compare against measured values and make it possible to identify where the fault is. Without the documentation there is nothing to compare to.

**Q7. Why should DC polarity be checked before connecting sensors and actuators?**

Many DC sensors, actuators and PLC field devices are polarity-sensitive. They are designed to operate with the positive supply rail connected to their positive (V+) terminal and the negative or common rail connected

to their negative (GND or V-) terminal. Reversing the polarity can permanently destroy the internal electronics of the device. For example, a 4-20 mA pressure transmitter or a transistor output module connected in reverse polarity will fail immediately when power is applied. Checking polarity before connection prevents this irreversible damage.

**Q8. What does the term 'common reference' mean in a low-voltage control circuit?**

The common reference (also called the common, the 0 V rail, or the negative rail) is the shared electrical return path that all voltages in the circuit are measured against. In a 24 V DC control circuit, the 0 V terminal of the supply is the common reference. When we say a sensor outputs 5 V, we mean 5 V measured between the sensor output and the common reference, not between two arbitrary points. All devices in the circuit — the PLC, the sensors, the actuators — must connect their 0 V or GND terminals to the same common reference rail. Without a common reference, voltage measurements are meaningless and signal circuits will not work correctly.

**Q9. State one reason why water and metal jewellery are not allowed around circuit work.**

Water is an electrically conductive liquid. If water (or any liquid) contacts the circuit board or supply terminals, it creates an unintended conductive path between terminals and components. This can cause a short circuit, destroy components, corrupt sensor readings, or in the case of higher voltages, cause an electric shock. Metal jewellery such as rings, bracelets or watches is similarly conductive. If a ring or bracelet bridges two energised terminals (for example, across a battery or supply output) it will carry a large short-circuit current, causing rapid and severe heating of the metal which can result in a serious burn injury to the wearer.

**Q10. Write one action you must take when you smell burning insulation or see smoke from a circuit.**

Immediately switch off the power supply using the main power switch or isolator. Do not touch the circuit, the components, or any wire until the supply is confirmed off and the circuit has had time to cool. Inform the instructor or lab supervisor immediately and do not attempt to restart or investigate the circuit alone. Only resume work after the instructor has inspected the circuit, identified the cause of the burning, and given clearance to proceed.



## Conclusion

---

This Lab Safety Readiness Pack has demonstrated understanding of the ten core safety competencies required before beginning any circuit work in ACICP 101. The hazard identification table in Part B showed that every electrical scenario — from a direct short circuit to polarity reversal and supply overloading — carries a clear consequence that can be prevented by a simple, calculated control measure. The pre-lab quiz in Part C confirmed ability to apply Ohm's Law ( $I = V / R$  giving 200 mA for a 120 ohm relay coil on a 24 V supply), to distinguish between short and open circuit faults, to explain the roles of fuses, polarity checking and common reference in safe circuit operation, and to respond correctly to emergency conditions such as smoke or burning. These safety habits will be applied in every simulation and supervised practical activity throughout the ACICP 101 course.

## Student Declaration

---

I confirm that this submission is my own work and that all sources, screenshots, simulations, and calculations used have been honestly presented.

I understand that unsafe electrical practice, copied work, or fabricated screenshots may attract disciplinary action.

|                             |                                                                                   |
|-----------------------------|-----------------------------------------------------------------------------------|
| <b>Student Name:</b>        | <i>SUNUSI ABDULLAHI</i>                                                           |
| <b>Registration Number:</b> | <i>C11-26-ACIS-17362</i>                                                          |
| <b>Signature:</b>           |  |
| <b>Date:</b>                | 02/JULY/2026                                                                      |